Task-Weighted Observable Subspaces for Repeated Multi-Angle QAOA

Molena Huynh
North Carolina State University
molena.huynh@jmp.com

Abstract

Repeated optimization on a graph changes task weights while preserving the circuit and its observable geometry. We formulate this setting as task-weighted dimension reduction for multi-angle QAOA. For observable Jacobian $\mathbf{J}$ and task second moment $\mathbf{M} = \mathbb{E}[ww^\intercal] = LL^\intercal$, the leading singular subspace of $\mathbf{J}^\intercal L$ minimizes expected discarded local gradient energy. A randomized algorithm constructs the subspace from Jacobian-vector and vector-Jacobian actions; a residual diagnostic governs refreshes. We audit scalar objectives, vector observables, shots, checks, refreshes, and simulator derivatives separately. Across 16 graph-depth cells, residual-gated reuse changes mean approximation ratio by -1.57 ± 1.04 percentage points relative to full-space SPSA while using 1.218 times as many circuit evaluations, so the criterion fails. A 256-shot objective pilot favors per-task refresh by $+0.61 \pm 1.57$ points but retains exact simulator-assisted construction. The evidence establishes the local representation and its reproducible cost ledger, not query, hardware, scaling, or quantum advantage.

1 Introduction

The quantum approximate optimization algorithm (QAOA) alternates problem-separating and mixing unitaries before optimizing their angles with a classical routine [22, 23]. It is a canonical variational quantum algorithm because every outer-loop decision changes the number of circuit preparations required to reach a useful objective value [42, 47]. This interaction is particularly important in the noisy intermediate-scale regime, where circuit depth, sampling noise, and classical search all constrain practical evidence [10, 49].

Standard QAOA uses two angles per layer. Multi-angle QAOA assigns separate phase angles to cost terms and separate mixer angles to vertices, yielding $d = p(|E|+|V|)$ coordinates at depth p [27]. The larger parameterization can increase expressivity, but a high-dimensional parameter vector need not imply that every direction changes the observables relevant to a task [14, 66]. Concentration and transfer phenomena further suggest that useful structure can recur across related instances or objectives [2, 12].

We study a repeated-task regime on a fixed graph. Let $F(\boldsymbol{\theta}) \in \mathbb{R}^m$ collect the expected edge-cut observables and let a task be $C_w(\boldsymbol{\theta}) = w^\intercal F(\boldsymbol{\theta})$. Only w changes between tasks. Weighted-MaxCut transfer and warm-start studies motivate reuse, but do not identify which local parameter directions preserve a distribution of weighted objectives [20, 54]. Graph-based initialization methods similarly transfer angle information without solving this observable-space problem [32].

The distinction matters because a scalar objective can hide the physical and statistical cost of representation learning. A single reverse-mode vector-Jacobian product returns $\mathbf{J}^\intercal w$ on a differentiable simulator, whereas a device must estimate observables through measurements or parameter-shift evaluations [52, 62]. Simultaneous perturbation stochastic approximation (SPSA) uses two scalar objective evaluations per update in either the full or reduced space [56]. Consequently, a lower optimized dimension does not automatically reduce the number of quantum queries.

Optimization can also become intrinsically difficult when gradients concentrate near zero [43]. Expressibility and control structure affect this phenomenon [29, 36], and gradient-free optimization does not remove it [4]. Any proposed reduction must therefore separate a mathematical statement about retained local gradient energy from an empirical statement about end-to-end optimization.

Our construction weights the observable Jacobian by the second moment of a training task distribution. It then applies fixed-precision randomized range finding without forming the dense Jacobian [25, 41]. The implementation follows blocked randomized QB ideas [65], but its target matrix is the task-weighted parameter sensitivity $A = \mathbf{J}^\intercal L$ rather than a data matrix.

Figure 1 summarizes the separation between basis-training weights, evaluation tasks, local construction, and the matched audit. The central conclusion is negative but informative: the task-weighted subspace is well defined and empirically better than a random basis, yet the tested refresh policies do not establish an operational advantage.

Our contributions are fourfold. First, we prove the exact optimality criterion for task-weighted local subspaces. Second, we give a matrix-free construction and a complete resource ledger. Third, we distinguish reusable geometry from objective-conditioned refresh and compare both against matched controls. Fourth, we publish a de-

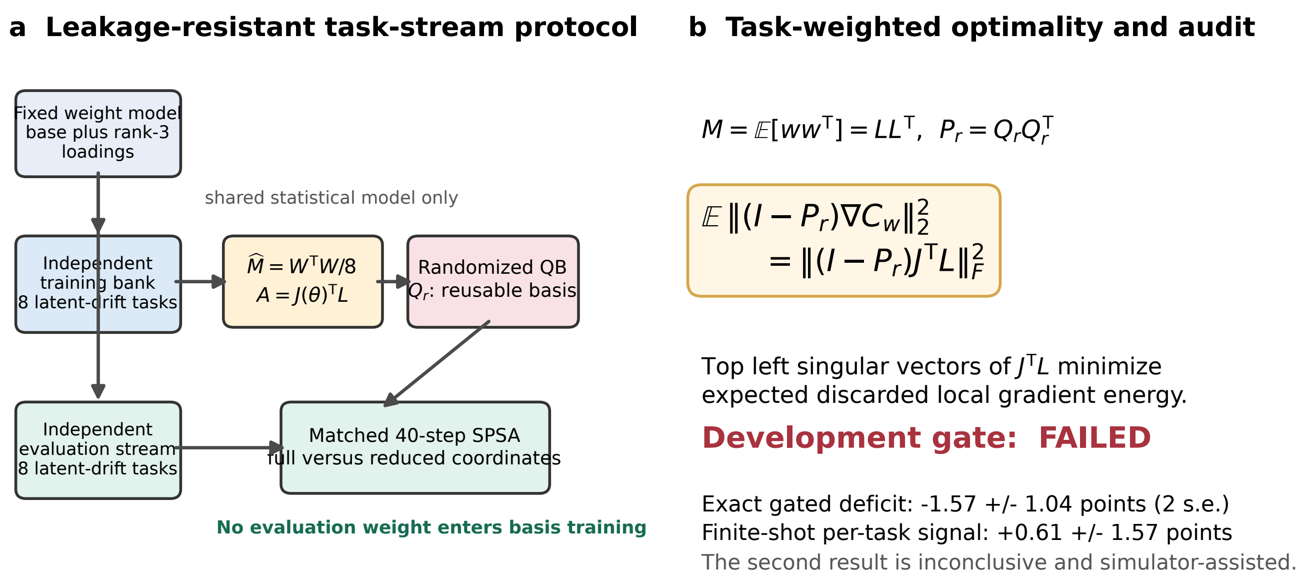

Figure 1: **Leakage-resistant repeated-task protocol.** A training bank defines the empirical task second moment; an independent evaluation stream tests transfer. Randomized QB constructs a local basis from $\mathbf{J}^\top L$, and full and reduced SPSA receive the same 40-step schedule. The displayed development decision is generated from the committed evidence.

terministic reproduction chain: numerical displays are regenerated from frozen CSV and JSON records, while protocol schematics are rendered deterministically from committed configuration metadata.

2 Background

2.1 Multi-Angle QAOA

For an undirected graph $G = (V, E)$ with $n = |V|$ and $m = |E|$, define the edge operator $C_e = (I - Z_i Z_j)/2$ for $e = (i, j)$. A depth-p multi-angle state is

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{\ell=1}^{p} \left[\prod_{i\in V} e^{-i\beta_{\ell i} X_i} \prod_{e\in E} e^{-i\gamma_{\ell e} C_e} \right] |+\rangle^{\otimes n}. \quad (1)$$

The parameter vector contains $p(m+n)$ angles. The vector observable and its Jacobian are

$$F(\boldsymbol{\theta}) = \left(\langle\psi(\boldsymbol{\theta})|C_e|\psi(\boldsymbol{\theta})\rangle\right)_{e\in E}, \qquad \mathbf{J}(\boldsymbol{\theta}) = \frac{\partial F}{\partial \boldsymbol{\theta}} \in \mathbb{R}^{m\times d}. \quad (2)$$

Analytic differentiation can evaluate individual parameter derivatives [40]; quantum-natural-gradient methods instead change the metric in which an update is chosen [57]. Our object is different: it is a low-dimensional Euclidean subspace of parameters chosen to retain a distribution of scalarized observable gradients.

All MaxCut edge terms commute in the computational basis. One bitstring therefore updates every component of F, and many task scalarizations may be computed from the same measurement batch. This form of observable reuse is related in spirit to predicting many properties from shared measurements [30], but it does not require general shadow tomography [1].

2.2 Repeated Weighted Objectives

Let $w \in \mathbb{R}^m$ be a random task weight with finite second moment. Choose an integer $q \geq 1$ and $L \in \mathbb{R}^{m\times q}$ such that

$$\mathbf{M} = \mathbb{E}[ww^\top] = LL^\top. \quad (3)$$

The scalar objective $C_w(\boldsymbol{\theta}) = w^\top F(\boldsymbol{\theta})$ has local gradient

$$\nabla C_w(\boldsymbol{\theta}) = \mathbf{J}(\boldsymbol{\theta})^\top w. \quad (4)$$

If $Q \in \mathbb{R}^{d\times r}$ is orthonormal and $P = QQ^\top$, the quantity

$$\mathcal{L}(P) = \mathbb{E}_w \left\|(\mathbf{I} - P)\nabla C_w(\boldsymbol{\theta})\right\|_2^2 \quad (5)$$

measures discarded local gradient energy. It does not measure global suboptimality, and it changes when the anchor $\boldsymbol{\theta}$ moves.

Active-subspace methods also average gradient outer products [16]. Sufficient-dimension-reduction methods, including sliced inverse regression, solve related but statistically distinct projection problems [37]. Regression graphics provides a broader account of response-preserving low-dimensional views [17]. Here, the vector-observable factorization exposes the projection target exactly:

$$\mathcal{L}(P) = \left\|(\mathbf{I} - P)\mathbf{J}^\top L\right\|_F^2. \quad (6)$$

2.3 Randomized Range Approximation

Set $A = \mathbf{J}^\mathsf{T} L \in \mathbb{R}^{d\times q}$. A randomized range finder probes $A\Omega$ and orthogonalizes the result; a blocked fixed-precision algorithm adds directions until a residual estimator falls below tolerance [63]. Deterministic streaming sketches offer another route when rows arrive sequentially [38]. Concentration inequalities justify probabilistic residual controls under explicit random matrix assumptions [58], but the one-probe operational refresh rule used here is not promoted to such a bound.

The matrix is accessed through

$$Av = \mathbf{J}^\mathsf{T}(Lv), \qquad A^\mathsf{T} u = L^\mathsf{T}(\mathbf{J}u). \tag{7}$$

The first action is a vector-Jacobian product and the second is a Jacobian-vector product. Our exact-statevector implementation uses reverse automatic differentiation for the former and central differences for the latter.

3 Related Work

Parameter concentration and interpolation can reduce the search needed by QAOA without learning a new local basis [2, 66]. Warm starts incorporate classical solutions [20], while learned optimizers predict search behavior from training instances [34]. Broad optimizer studies show that rankings depend on budget, noise, and objective geometry [11, 44].

Graph representation learning provides invariant or equivariant features for such transfer. Graph convolutional networks aggregate normalized neighborhoods [35]; graph attention assigns learned neighbor weights [59]; and graph-isomorphism networks clarify the discriminative power of message passing [64]. Our basis is not a graph embedding. It is constructed within one graph from circuit sensitivities and a task distribution.

The outer-loop optimization literature supplies several relevant controls. Mesh-adaptive direct search is derivative free with a convergence framework [5]; BOBYQA builds a bounded quadratic interpolation model [48]; and covariance-matrix adaptation learns a search distribution [26]. These methods alter the search strategy, whereas RSQ fixes the optimizer and changes only its coordinate subspace.

Bayesian optimization models an expensive response surface. Efficient global optimization introduced an improvement-based acquisition rule [33]; practical machine-learning variants made this framework widely usable [55]; and modern reviews organize its modeling and acquisition choices [53]. Batched Monte Carlo acquisition functions can be implemented with differentiable software [6]. None of these eliminates the need to count the quantum evaluations used to fit the surrogate.

Repeated tuning also resembles algorithm configuration. Sequential model-based configuration chooses settings from prior runs [31], while SMAC exposes a reproducible implementation of this approach [39]. The classical algorithm-selection problem is older and more general [50]. It motivates our use of disjoint training and evaluation streams, because reusing the same tasks for basis choice and performance claims invites selection bias [13].

Finally, the graph families in our experiments are controlled benchmarks: Erdős–Rényi random graphs [21], preferential-attachment graphs [7], and small-world constructions [61]. They are generated with NetworkX [24]. Results on these small synthetic families do not establish asymptotic or application-level generalization.

4 Shortcomings of Static and Unweighted Subspaces

4.1 Inconsistency under Task Shift

A static rank-r subspace may preserve gradients for the training task distribution and discard them under a shifted distribution. The issue is algebraic, not an artifact of randomized approximation.

Theorem 4.1 (A shifted task can be completely discarded)**.** *Fix an anchor with Jacobian $\mathbf{J} \in \mathbb{R}^{m\times d}$ and an orthogonal projector P of rank r. If $\operatorname{rank}(\mathbf{J}) > r$, then there exists a nonzero task weight $w_\star \in \mathbb{R}^m$ such that*

$$P\mathbf{J}^\mathsf{T} w_\star = 0 \quad \textit{and} \quad \mathbf{J}^\mathsf{T} w_\star \neq 0.$$

Consequently, a distribution concentrated at $w_\star$ has unit relative discarded gradient energy for P.

The full proof is in Appendix A.1. The theorem does not say that every shift is adversarial. It says that no rank-deficient local representation can be distribution free when the observable Jacobian has larger rank. Its construction permits signed scalarizations and therefore does not, by itself, establish complete loss for task weights restricted to the nonnegative orthant. Monitoring principal-angle drift or retained energy is necessary when the task law changes.

Classical statistical safeguards cannot repair an omitted physical direction after the fact. Bootstrap intervals describe sampling variation [19]; exact binomial limits quantify event rates [15]; and distribution-free inequalities can bound bounded averages [28]. Each requires a sampling model that is separate from the representation question.

4.2 Refresh Bias and Hidden Physical Cost

The residual diagnostic uses Gaussian directions v_j and estimates

$$\widehat{\rho}^2 = \frac{\sum_{j=1}^{s} [D_{(\mathbf{I}-P)v_j} C_w(\boldsymbol{\theta})]^2}{\sum_{j=1}^{s} [D_{v_j} C_w(\boldsymbol{\theta})]^2}. \tag{8}$$

Its numerator and denominator estimate squared directional energies as the finite-difference step vanishes, but

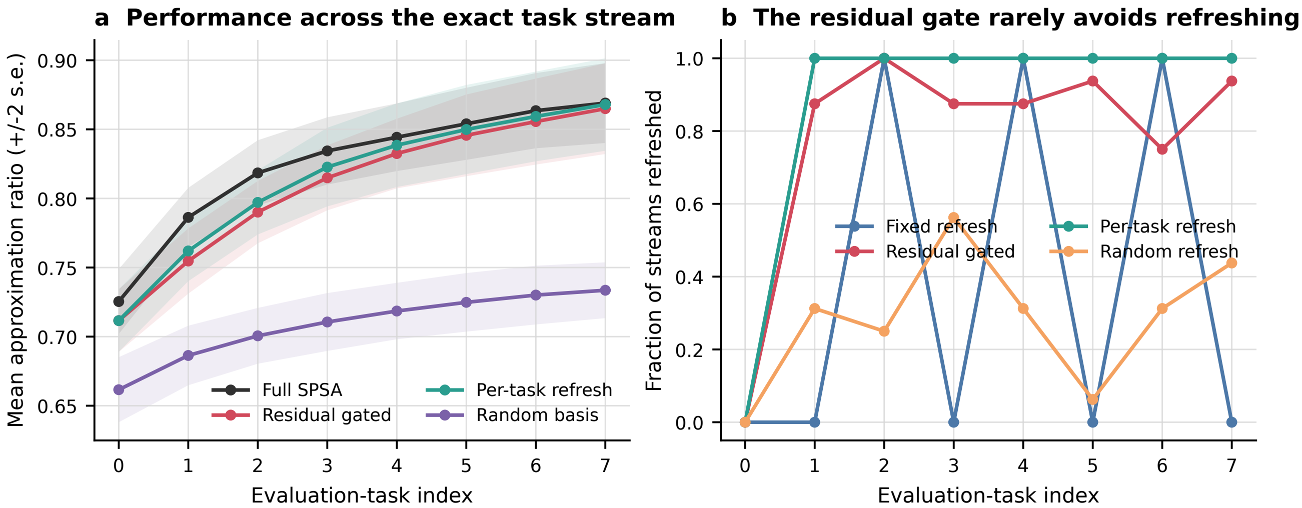

Figure 2: **Evaluation-task trajectories.** The left panel reports task-index performance after nesting repeated measurements within a graph-depth cell. The right panel shows that residual gating refreshes on most eligible transitions, limiting amortization in the tested regime.

their ratio is biased and the tested $s = 1$ rule is not a calibrated confidence statement. Conformal prediction offers distribution-free marginal coverage under exchangeability [3, 60]; conformalized quantile regression adapts interval width to covariates [51]; and extensions handle some nonexchangeable settings [8]. Those results do not apply automatically to adaptively refreshed quantum trajectories.

Cost accounting yields a second limitation. Suppose a coordinate-gradient method uses $2d$ circuit points per full update and $2r$ per reduced update, there are S updates per task and K tasks, and basis construction plus refreshes cost $C_{\text{basis}}(K)$ circuit points across those tasks.

Proposition 4.2 (Coordinate-gradient break-even condition)**.** *Under the preceding ledger, let S, K be positive integers, let $0 \le r < d$, and let $C_{\text{basis}}(K) \ge 0$. The reduced method uses fewer circuit points than the full method if and only if*

$$C_{\text{basis}}(K) < 2S(d - r)K. \tag{9}$$

If $C_{\text{basis}}(K) = C_{\text{basis}}$ is independent of K, this is equivalent to

$$K > \frac{C_{\text{basis}}}{2S(d - r)}. \tag{10}$$

Appendix A.2 proves the equivalence. It does not apply to SPSA, because both coordinate systems use two perturbed objectives per update. Nor may simulator VJPs be silently converted into hardware queries.

4.3 No Scalar-Query Saving under Matched SPSA

The preceding coordinate-gradient calculation does not transfer to an optimizer whose query count is dimension independent. For matched SPSA, a subspace may change the search geometry, but it cannot reduce the two scalar objectives used by each update.

Proposition 4.3 (SPSA accounting obstruction)**.** *Suppose full-space and rank-r SPSA each use S updates and one final scalar objective per task for K tasks. If the reduced method incurs nonnegative basis, diagnostic, and refresh cost C_{basis} measured in additional forward circuit points, then*

$$\begin{aligned} C_{\text{full}} &= (2S + 1)K, \\ C_{\text{reduced}} &= (2S + 1)K + C_{\text{basis}} \ge C_{\text{full}}. \end{aligned} \tag{11}$$

Equality holds if and only if $C_{\text{basis}} = 0$.

Appendix A.3 proves the statement from the SPSA oracle ledger. Thus any end-to-end saving in the present access model would need fewer updates, cheaper per-objective measurement, or amortization over some other resource; parameter compression alone is insufficient under the matched stochastic-gradient and hardware-derivative access models considered here [52, 56, 62].

5 Task-Weighted Observable Subspaces

5.1 Optimal Task-Weighted Subspace

Theorem 5.1 (Optimal local parameter subspace)**.** *Let $w \in \mathbb{R}^m$ be square-integrable with $\mathbf{M} = \mathbb{E}[ww^\mathsf{T}] = LL^\mathsf{T}$. Fix a differentiable anchor with Jacobian $\mathbf{J} \in \mathbb{R}^{m \times d}$, set $A = \mathbf{J}^\mathsf{T} L$, and fix $0 \le r \le d$. For every rank-r orthogonal projector P,*

$$\mathbb{E} \left\| (\mathbf{I} - P)\mathbf{J}^\mathsf{T} w \right\|_2^2 = \|(\mathbf{I} - P)A\|_F^2 . \tag{12}$$

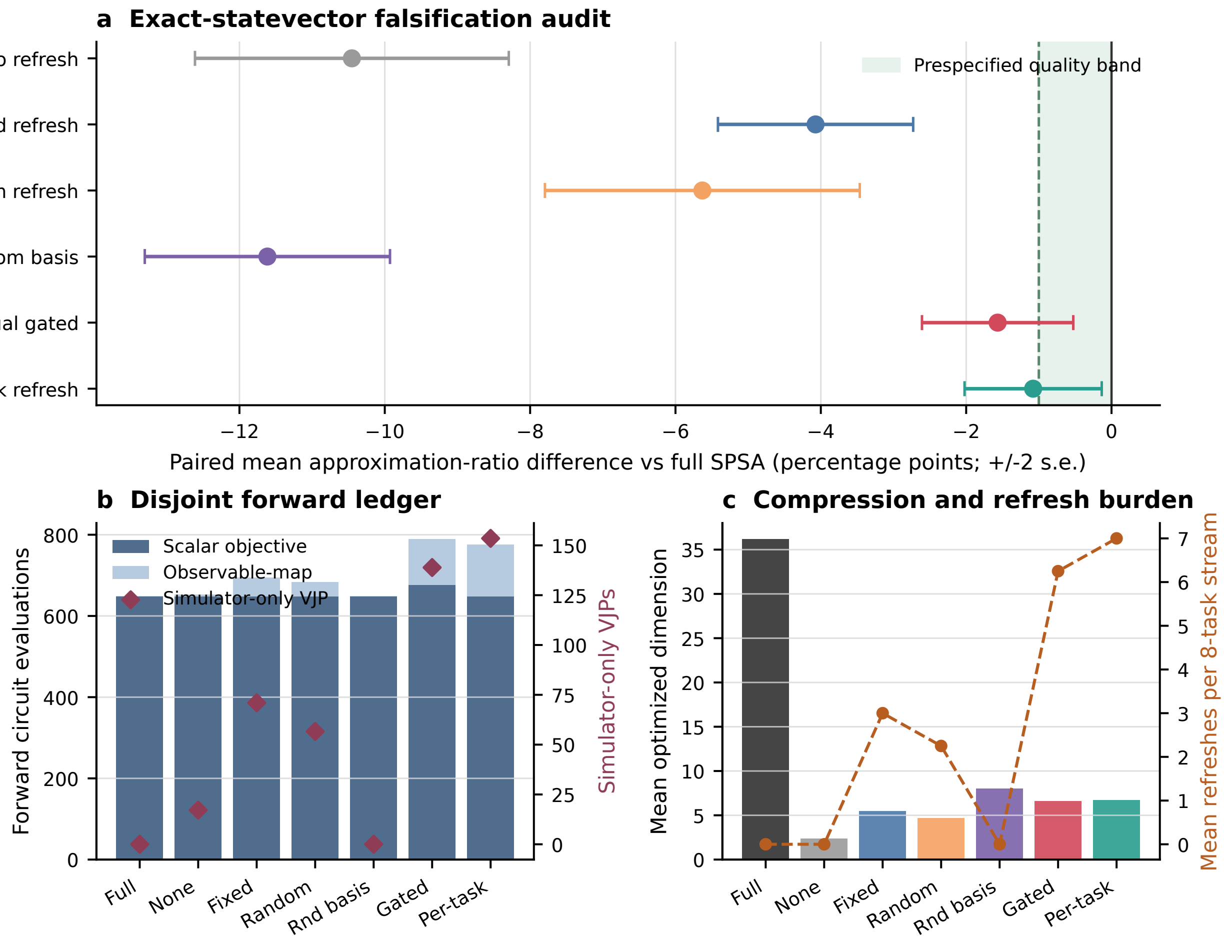

Figure 3: **Exact-statevector falsification audit.** Residual-gated reuse misses the declared one-percentage-point quality band and spends additional observable evaluations and simulator VJPs. Error bars are descriptive ± 2 standard errors over configured graph-depth cells.

If $A = U\Sigma V^{\top}$ is a full singular-value decomposition with $U \in \mathbb{R}^{d\times d}$, then $P_r = U_r U_r^{\top}$ minimizes this loss over all rank-r orthogonal projectors, and the minimum equals $\sum_{j>r} \sigma_j(A)^2$.

Appendix A.4 gives a complete trace and variational proof. The result is an exact population statement. With training matrix $\mathbf{W} \in \mathbb{R}^{K_{\mathrm{tr}}\times m}$, the implementation substitutes

$$\widehat{\mathbf{M}} = K_{\mathrm{tr}}^{-1}\mathbf{W}^{\top}\mathbf{W}, \qquad L = K_{\mathrm{tr}}^{-1/2}\mathbf{W}^{\top}, \tag{13}$$

so the theorem holds exactly for the empirical task distribution.

The result is a local analogue of best low-rank approximation, but the weighting is essential. Unweighted observable sensitivity gives equal importance to task directions that may never occur. For $0 < r < d$, the top singular space is unique only when $\sigma_r(A) > \sigma_{r+1}(A)$; otherwise every projector onto an admissible leading invariant subspace attains the same loss.

5.2 Matrix-Free Construction

Blocked randomized QB starts from a Gaussian test block, applies the two actions in Eq. (7), orthogonalizes against the accumulated basis, and repeats until a relative residual condition or maximum rank is reached. Recycling initializes a new build from the previous basis but allows rotation. The method stores every stop reason, residual, rank, and operator count.

The two-sided implementation uses simulator VJPs and finite-difference JVPs. The forward-only control replaces VJPs by additional finite differences. A dense-SVD oracle materializes $\mathbf{J}$ and is reported separately rather than treated as a deployable method. This separation follows the broader principle that randomized approximation quality and operator-access cost must be reported together [25, 63].

5.3 Residual-Gated Refresh

The training task bank and evaluation stream share only a fixed latent weight model. Their random seeds and realized weights are disjoint. At each transition, one of six policies is applied: no refresh, fixed refresh, residual-gated refresh, per-task refresh, random refresh, or a fixed isotropic random basis. The random-basis control isolates low dimension from learned geometry; fixed and random refresh controls separate timing from content.

Multiple diagnostic comparisons are descriptive. False-discovery-rate procedures address families of statistical tests [9], and Neyman–Pearson testing formalizes simple-hypothesis power [45]. We do not retrofit either framework after examining the development grid. The only decision is the stated three-part development gate.

5.4 Matched Full and Reduced SPSA

For native parameters or reduced coordinates z with $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + Qz$, SPSA draws a Rademacher perturbation δ_k:

$$\widehat{g}_k = \frac{C(x_k + c_k\delta_k) - C(x_k - c_k\delta_k)}{2c_k}\delta_k. \tag{14}$$

Both matched SPSA update loops use 40 updates, identical gain schedules, one perturbation pair per update, and one final objective, for 81 optimization-objective calls per task. Residual diagnostics add separately counted scalar calls to reduced methods. The comparison therefore does not claim that reduced SPSA saves objective queries. Derivative-free methods can respond differently to dimension and noise [5, 26]; the matched design holds the optimizer family fixed to isolate the representation.

6 Experiments

6.1 Exact-Statevector Repeated-Task Benchmark

The exact grid contains random 3-regular and Erdős–Rényi graphs with $n \in \{8, 10\}$, two graph seeds, and $p \in \{1, 2\}$. Eight topologies yield 16 configured topology-depth cells. Each graph receives eight training weights and an independent eight-task evaluation trajectory generated by a positive rank-three latent model with autoregressive drift. Weight vectors are normalized to mean one.

The statevector backend uses complex-128 PyTorch tensors [46]. Brute-force weighted MaxCut supplies the denominator of each approximation ratio. Results average tasks within a graph-depth cell; paired differences subtract full SPSA in the same cell. Since both depths reuse each topology, the reported ± 2 standard errors are descriptive and do not model cross-depth dependence. Comparisons across multiple data sets require care even under independent sampling [18].

6.2 Matched Development Audit

The development gate requires the residual-gated method to be no more than one approximation-ratio point below full SPSA, to exceed the random basis, and to use at most 1.25 times the forward circuit evaluations of full SPSA. The gated method achieves a paired difference of -1.57 ± 1.04 points and a cost ratio of 1.218. It beats the random basis but fails the quality condition. Per-task refresh gives -1.08 ± 0.94 points and uses still more construction. The negative decision is not changed by selecting the most favorable component.

Hyperparameter selection and final evaluation on the same grid can exaggerate performance [13]. We therefore treat this experiment as development only. Standard bootstrap methodology could quantify resampling uncertainty [19], but the small dependent grid would still require an explicit topology-level sampling model.

6.3 Finite-Shot Pilot

The pilot retains $p = 2$ and uses eight topology cells. Each scalar objective is estimated from 256 computational-basis shots, with three measurement repeats nested inside each method-cell summary. Candidate selection uses the sampled objective, while the reported approximation ratio is the exact expectation of the selected parameters. This reduces reporting noise but further separates the study from a hardware-only protocol.

Residual-gated reuse changes the mean by -0.33 ± 1.67 points and per-task refresh by $+0.61 \pm 1.57$ points relative to full SPSA. Neither interval resolves a stable advantage. Exact binomial reasoning would be appropriate for binary events under independent trials [15]; it is not a substitute for the continuous, paired, repeated-measurement analysis used here.

The experiment supports three conclusions. First, task weighting identifies a nonrandom local representation. Second, refresh frequency is the dominant obstacle to amortization on the tested short streams. Third, the current simulator-assisted construction cannot support a hardware-efficiency claim. Longer task banks, larger graphs, calibrated refresh decisions, and device-native sensitivity actions remain untested.

7 Conclusion

We derived a task-weighted observable tangent representation for repeated multi-angle QAOA and proved its exact local optimality. Matrix-free randomized QB avoids dense Jacobian materialization and exposes every sensitivity action. The representation is mathematically useful, but the matched evidence rejects the tested operational claim: residual-gated reuse misses the development quality threshold, refreshes frequently, and spends additional forward evaluations.

Limitations. The study has five principal limitations. It uses at most ten qubits and depth two; its graph

Table 1: **Legacy single-objective benchmark.** Values are regenerated from `experiments/results/maxcut_small.csv`. This track tests local compression on independent objectives and is not merged with the repeated-task development decision.

Method	Depth	Parameters	Approx. ratio	Paired change
Full ma-QAOA	1	23.7	0.862 ± 0.053	–
Symmetry-tied	1	14.4	0.845 ± 0.065	-1.67 ± 1.42 pp
Fixed-rank oracle	1	8.0	0.764 ± 0.051	-9.77 ± 2.09 pp
RSQ ($\varepsilon = 0.01$)	1	14.7	0.867 ± 0.051	$+0.48 \pm 0.94$ pp
RSQ ($\varepsilon = 0.1$)	1	14.7	0.862 ± 0.051	-0.03 ± 0.79 pp
Forward-only RSQ	1	8.0	0.798 ± 0.055	-6.39 ± 1.98 pp
Full ma-QAOA	2	47.4	0.945 ± 0.044	–
Symmetry-tied	2	28.7	0.940 ± 0.064	-0.45 ± 2.64 pp
Fixed-rank oracle	2	8.0	0.764 ± 0.048	-18.07 ± 2.54 pp
RSQ ($\varepsilon = 0.01$)	2	14.7	0.897 ± 0.054	-4.77 ± 2.47 pp
RSQ ($\varepsilon = 0.1$)	2	14.7	0.906 ± 0.058	-3.88 ± 1.84 pp
Forward-only RSQ	2	8.0	0.797 ± 0.053	-14.73 ± 2.65 pp

Table 2: **Exact repeated-task audit.** Objective and observable evaluations are disjoint forward categories. VJPs are simulator-only operations and are not added to the forward total.

Method	Ratio ± 2 s.e.	Δ full (points)	Dim.	Obj.	Obs.	VJP	Refresh
Full SPSA	0.8244 ± 0.0230	0.00 (ref.)	36.2	648	0.0	0.0	0.00
No refresh	0.7199 ± 0.0238	-10.45 ± 2.16	2.4	648	4.8	17.1	0.00
Fixed refresh	0.7837 ± 0.0236	-4.07 ± 1.34	5.5	648	46.8	71.0	3.00
Residual gated	0.8087 ± 0.0237	-1.57 ± 1.04	6.6	676	113.2	138.9	6.25
Per-task refresh	0.8136 ± 0.0262	-1.08 ± 0.94	6.7	648	128.0	153.5	7.00
Random refresh	0.7681 ± 0.0258	-5.63 ± 2.17	4.7	648	34.8	56.6	2.25
Random basis	0.7083 ± 0.0202	-11.62 ± 1.69	8.0	648	0.0	0.0	0.00

families are synthetic; its task law is a controlled rank-three drift model; its uncertainty summaries are descriptive; and its subspace construction uses exact simulator derivatives. These limitations exclude claims of quantum advantage, hardware advantage, asymptotic scaling, or distribution-free transfer. General lessons about experiment design remain valid: comparisons should use matched optimizers, preserve failed gates, report strata, and keep simulator operations distinct from physical measurements.

A confirmatory design is included but deliberately unexecuted. It specifies 40 new topology units, longer task streams, changepoints, independent-weight stress tests, shot and readout perturbations, device-native coordinate gradients, and dense-oracle controls. Its decision rule must be frozen before the outcomes are generated. Distribution-free calibration is attractive only when its exchangeability conditions are defensible [3, 8]; otherwise the resulting coverage statement must be qualified.

The installable package exposes `rsqaoa-experiment` for a direct run and `rsqaoa-reproduce` for full evidence regeneration or validation. Source records, protocol hashes, implementation hashes, and generated-asset hashes are committed with the manuscript. This makes every numerical table and figure traceable without implying that deterministic rerunning removes modeling uncertainty.

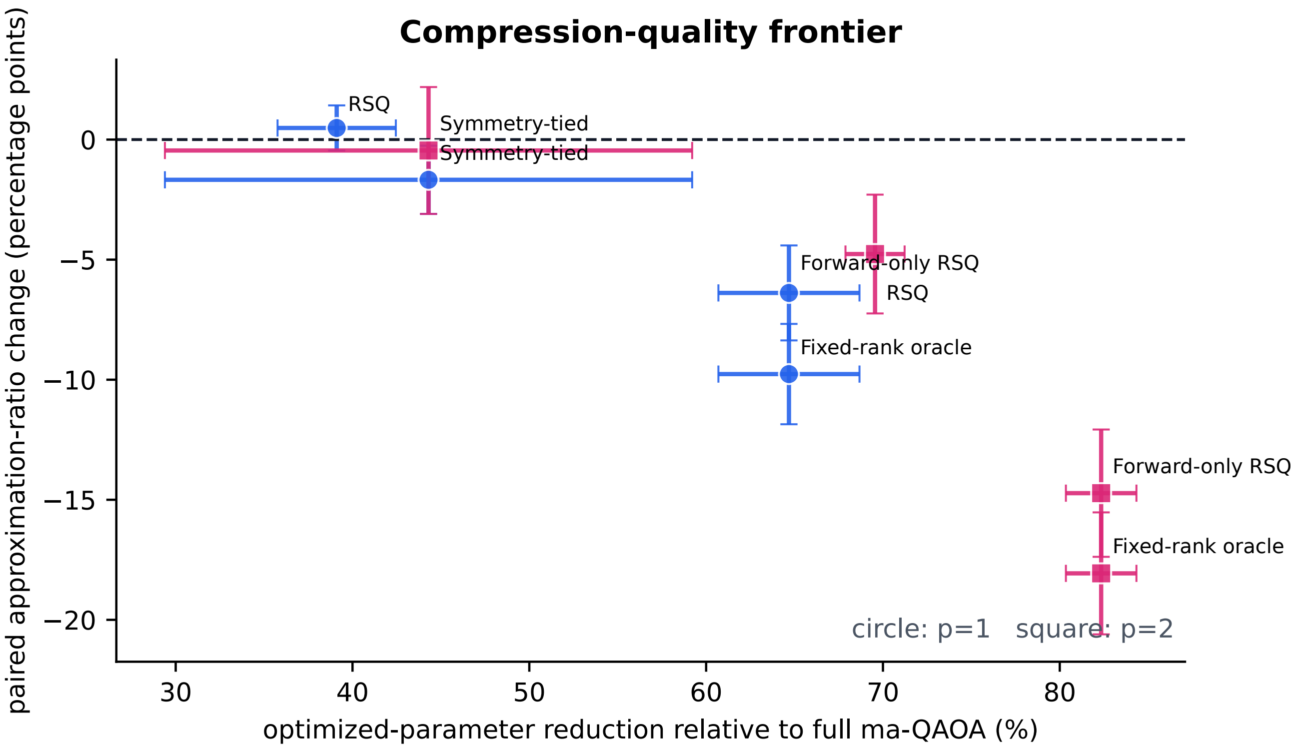

Figure 4: **Legacy compression-quality tradeoff.** The figure preserves the original exact-statevector evidence track and exposes the stronger depth dependence that motivated the repeated-task audit.

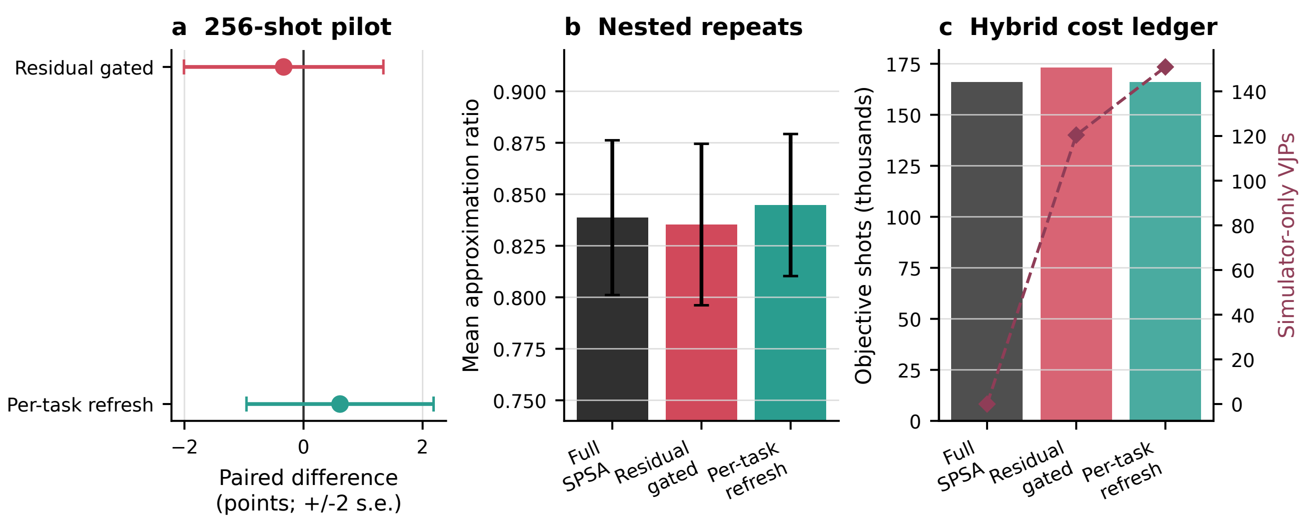

Figure 5: **Finite-shot objective pilot.** Scalar objectives use 256 shots, while subspace construction remains exact and simulator-assisted. Consequently, the figure is not an end-to-end hardware experiment.

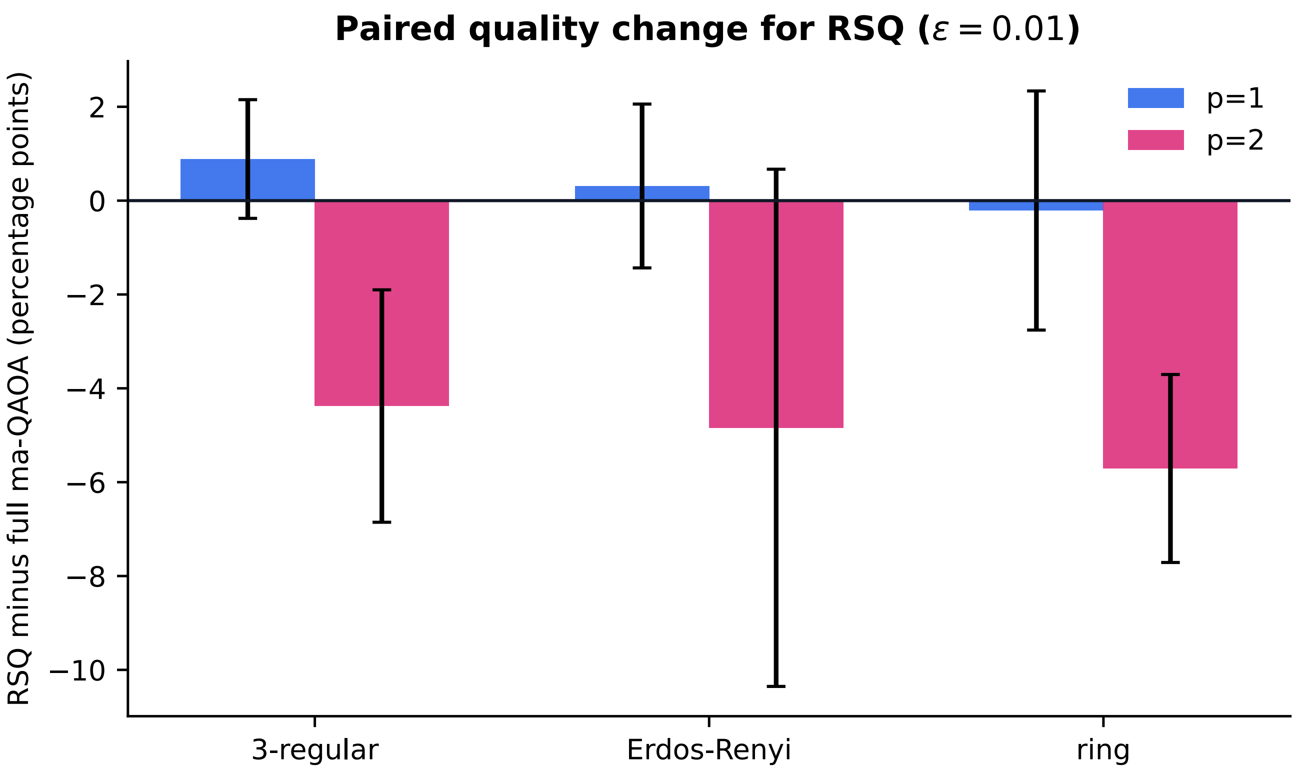

Figure 6: **Legacy family-stratified behavior.** Performance depends on graph family and depth, so pooled averages are accompanied by stratified diagnostics.

References

[1] Scott Aaronson. Shadow tomography of quantum states. In *Proceedings of the 50th Annual ACM SIGACT Symposium on Theory of Computing*, pages 325–338, 2018. doi: 10.1145/3188745.3188802.

[2] V. Akshay, D. Rabinovich, E. Campos, and J. Biamonte. Parameter concentrations in quantum approximate optimization. *Physical Review A*, 104:L010401, 2021. doi: 10.1103/PhysRevA.104.L010401.

[3] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv:2107.07511, 2021. URL https://arxiv.org/abs/2107.07511.

[4] Andrew Arrasmith, M. Cerezo, Piotr Czarnik, Lukasz Cincio, and Patrick J. Coles. Effect of barren plateaus on gradient-free optimization. *Quantum*, 5: 558, 2021. doi: 10.22331/q-2021-10-05-558.

[5] Charles Audet and J. E. Dennis. Mesh adaptive direct search algorithms for constrained optimization. *SIAM Journal on Optimization*, 17(1):188–217, 2006. doi: 10.1137/040603371.

[6] Maximilian Balandat, Brian Karrer, Daniel R. Jiang, Samuel Daulton, Benjamin Letham, Andrew Gordon Wilson, and Eytan Bakshy. BoTorch: A framework for efficient monte-carlo bayesian optimization. In *Advances in Neural Information Processing Systems*, volume 33, pages 21524–21538, 2020.

[7] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439): 509–512, 1999. doi: 10.1126/science.286.5439.509.

[8] Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. arXiv:2202.13415, 2022. URL https://arxiv.org/abs/2202.13415.

[9] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.

[10] Kishor Bharti, Alba Cervera-Lierta, Thi Ha Kyaw, Tobias Haug, Sumner Alperin-Lea, Abhinav Anand, Matthias Degroote, Hermanni Heimonen, Jakob S. Kottmann, Tim Menke, Wai-Keong Mok, Sukin Sim, Leong-Chuan Kwek, and Alán Aspuru-Guzik. Noisy intermediate-scale quantum algorithms. *Reviews of Modern Physics*, 94:015004, 2022. doi: 10.1103/RevModPhys.94.015004.

[11] Xavier Bonet-Monroig, Hao Wang, Diederick Vermetten, Bruno Senjean, Charles Moussa, Thomas Bäck, Vedran Dunjko, and Thomas E. O'Brien. Performance comparison of optimization methods on variational quantum algorithms. *Physical Review A*, 107:032407, 2023. doi: 10.1103/PhysRevA.107.032407.

[12] Fernando G. S. L. Brandão, Michael Broughton, Edward Farhi, Sam Gutmann, and Hartmut Neven. For fixed control parameters the quantum approximate optimization algorithm's objective function value concentrates for typical instances. arXiv:1812.04170, 2018. URL https://arxiv.org/abs/1812.04170.

[13] Gavin C. Cawley and Nicola L. C. Talbot. On overfitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

[14] M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021. doi: 10.1038/s42254-021-00348-9.

[15] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

[16] Paul G. Constantine. *Active Subspaces: Emerging Ideas for Dimension Reduction in Parameter Studies*. SIAM, Philadelphia, 2015. doi: 10.1137/1.9781611973860.

[17] R. Dennis Cook. *Regression Graphics: Ideas for Studying Regressions Through Graphics*. Wiley, 1998. doi: 10.1002/9780470316931.

[18] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

[19] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993. doi: 10.1201/9780429246593.

[20] Daniel J. Egger, Jakub Mareček, and Stefan Woerner. Warm-starting quantum optimization. *Quantum*, 5:479, 2021. doi: 10.22331/q-2021-06-17-479.

[21] Paul Erdős and Alfréd Rényi. On random graphs i. *Publicationes Mathematicae Debrecen*, 6:290–297, 1959.

[22] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm. arXiv:1411.4028, 2014. URL https://arxiv.org/abs/1411.4028.

[23] Stuart Hadfield, Zhihui Wang, Bryan O'Gorman, Eleanor G. Rieffel, Davide Venturelli, and Rupak Biswas. From the quantum approximate optimization algorithm to a quantum alternating operator ansatz. *Algorithms*, 12(2):34, 2019. doi: 10.3390/a12020034.

[24] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*, pages 11–15, 2008.

[25] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011. doi: 10.1137/090771806.

[26] Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001. doi: 10.1162/106365601750190398.

[27] Rebekah Herrman, Phillip C. Lotshaw, James Ostrowski, Travis S. Humble, and George Siopsis. Multi-angle quantum approximate optimization algorithm. *Scientific Reports*, 12:6781, 2022. doi: 10.1038/s41598-022-10555-8.

[28] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

[29] Zoë Holmes, Kunal Sharma, M. Cerezo, and Patrick J. Coles. Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum*, 3:010313, 2022. doi: 10.1103/PRXQuantum.3.010313.

[30] Hsin-Yuan Huang, Richard Kueng, and John Preskill. Predicting many properties of a quantum system from very few measurements. *Nature Physics*, 16:1050–1057, 2020. doi: 10.1038/s41567-020-0932-7.

[31] Frank Hutter, Holger H. Hoos, and Kevin Leyton-Brown. Sequential model-based optimization for general algorithm configuration. In *Learning and Intelligent Optimization*, volume 6683, pages 507–523, 2011. doi: 10.1007/978-3-642-25566-3_40.

[32] Nishant Jain, Brian Coyle, Elham Kashefi, and Niraj Kumar. Graph neural network initialisation of quantum approximate optimisation. *Quantum*, 6:861, 2022. doi: 10.22331/q-2022-11-17-861.

[33] Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13:455–492, 1998. doi: 10.1023/A:1008306431147.

[34] Sami Khairy, Ruslan Shaydulin, Lukasz Cincio, Yuri Alexeev, and Prasanna Balaprakash. Learning to optimize variational quantum circuits to solve combinatorial problems. In *Proceedings of the AAAI Conference*, volume 34, pages 2367–2375, 2020. doi: 10.1609/aaai.v34i03.5616.

[35] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.

[36] Martin Larocca, Piotr Czarnik, Kunal Sharma, Gopikrishnan Muraleedharan, Patrick J. Coles, and M. Cerezo. Diagnosing barren plateaus with tools from quantum optimal control. *Quantum*, 6:824, 2022. doi: 10.22331/q-2022-09-29-824.

[37] Ker-Chau Li. Sliced inverse regression for dimension reduction. *Journal of the American Statistical Association*, 86(414):316–327, 1991. doi: 10.1080/01621459.1991.10475035.

[38] Edo Liberty. Simple and deterministic matrix sketching. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 581–588, 2013. doi: 10.1145/2487575.2487623.

[39] Marius Lindauer, Katharina Eggensperger, Matthias Feurer, André Biedenkapp, Difan Deng, Carolin Benjamins, Tim Ruhkopf, René Sass, and

Frank Hutter. SMAC3: A versatile bayesian optimization package for hyperparameter optimization. *Journal of Machine Learning Research*, 23:1–9, 2022.

[40] Andrea Mari, Thomas R. Bromley, and Nathan Killoran. Estimating the gradient and higher-order derivatives on quantum hardware. *Physical Review A*, 103:012405, 2021. doi: 10.1103/PhysRevA.103.012405.

[41] Per-Gunnar Martinsson and Joel A. Tropp. Randomized numerical linear algebra: Foundations and algorithms. *Acta Numerica*, 29:403–572, 2020. doi: 10.1017/S0962492920000021.

[42] Jarrod R. McClean, Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik. The theory of variational hybrid quantum-classical algorithms. *New Journal of Physics*, 18:023023, 2016. doi: 10.1088/1367-2630/18/2/023023.

[43] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018. doi: 10.1038/s41467-018-07090-4.

[44] J. A. Montañez-Barrera and K. Michielsen. Toward a linear-ramp QAOA protocol: Evidence of a scaling advantage in solving some combinatorial optimization problems. *npj Quantum Information*, 11:131, 2025. doi: 10.1038/s41534-025-01082-1.

[45] Jerzy Neyman and Egon S. Pearson. On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society A*, 231:289–337, 1933. doi: 10.1098/rsta.1933.0009.

[46] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[47] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O'Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5:4213, 2014. doi: 10.1038/ncomms5213.

[48] Michael J. D. Powell. The BOBYQA algorithm for bound constrained optimization without derivatives. Technical Report DAMTP 2009/NA06, University of Cambridge, 2009.

[49] John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018. doi: 10.22331/q-2018-08-06-79.

[50] John R. Rice. The algorithm selection problem. *Advances in Computers*, 15:65–118, 1976. doi: 10.1016/S0065-2458(08)60520-3.

[51] Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[52] Maria Schuld, Ville Bergholm, Christian Gogolin, Josh Izaac, and Nathan Killoran. Evaluating analytic gradients on quantum hardware. *Physical Review A*, 99:032331, 2019. doi: 10.1103/PhysRevA.99.032331.

[53] Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016. doi: 10.1109/JPROC.2015.2494218.

[54] Ruslan Shaydulin, Phillip C. Lotshaw, Jeffrey Larson, James Ostrowski, and Travis S. Humble. Parameter transfer for quantum approximate optimization of weighted MaxCut. *ACM Transactions on Quantum Computing*, 4(3):19, 2023. doi: 10.1145/3584706.

[55] Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. Practical bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems*, volume 25, 2012.

[56] James C. Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992. doi: 10.1109/9.119632.

[57] James Stokes, Josh Izaac, Nathan Killoran, and Giuseppe Carleo. Quantum natural gradient. *Quantum*, 4:269, 2020. doi: 10.22331/q-2020-05-25-269.

[58] Joel A. Tropp. An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning*, 8(1–2):1–230, 2015. doi: 10.1561/2200000048.

[59] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018.

[60] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.

[61] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998. doi: 10.1038/30918.

[62] David Wierichs, Josh Izaac, Cody Wang, and Cedric Yen-Yu Lin. General parameter-shift rules for quantum gradients. *Quantum*, 6:677, 2022. doi: 10.22331/q-2022-03-30-677.

[63] David P. Woodruff. Sketching as a tool for numerical linear algebra. *Foundations and Trends in Theoretical Computer Science*, 10(1–2):1–157, 2014. doi: 10.1561/0400000060.

[64] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations*, 2019.

[65] Wenjian Yu, Yu Gu, and Yaohang Li. Efficient randomized algorithms for the fixed-precision low-rank matrix approximation. *SIAM Journal on Matrix Analysis and Applications*, 39(3):1339–1359, 2018. doi: 10.1137/17M1141977.

[66] Leo Zhou, Sheng-Tao Wang, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin. Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices. *Physical Review X*, 10:021067, 2020. doi: 10.1103/PhysRevX.10.021067.

A Proofs

A.1 Proof for Theorem 4.1

Proof. Let $\mathcal{R} = \mathrm{range}(\mathbf{J}^\top) \subseteq \mathbb{R}^d$. By assumption, $\dim \mathcal{R} = \mathrm{rank}(\mathbf{J}) > r$. The range of P has dimension r, so

$$\dim(\mathcal{R} \cap \ker P) \geq \dim \mathcal{R} + \dim \ker P - d = \mathrm{rank}(\mathbf{J}) - r > 0.$$

Choose a nonzero $g \in \mathcal{R} \cap \ker P$. Since $g \in \mathrm{range}(\mathbf{J}^\top)$, there exists $w_\star \in \mathbb{R}^m$ such that $\mathbf{J}^\top w_\star = g$. Because $g \neq 0$, this weight is not in $\ker(\mathbf{J}^\top)$. Because $g \in \ker P$, $P\mathbf{J}^\top w_\star = Pg = 0$, while $\mathbf{J}^\top w_\star = g \neq 0$.

Let the shifted task distribution place probability one on $w_\star$. Then

$$\frac{\mathbb{E}\left\|(\mathbf{I} - P)\mathbf{J}^\top w\right\|_2^2}{\mathbb{E}\left\|\mathbf{J}^\top w\right\|_2^2} = \frac{\|(\mathbf{I} - P)g\|_2^2}{\|g\|_2^2} = \frac{\|g\|_2^2}{\|g\|_2^2} = 1,$$

where $(\mathbf{I} - P)g = g$ because $Pg = 0$. This proves the claim. □

A.2 Proof for Proposition 4.2

Proof. Under the stated ledger, K tasks with S full coordinate-gradient updates use

$$C_{\text{full}} = 2dSK$$

circuit points. The reduced procedure uses $2r$ points per update plus the complete basis and refresh cost:

$$C_{\text{reduced}} = C_{\text{basis}}(K) + 2rSK.$$

Thus $C_{\text{reduced}} < C_{\text{full}}$ is equivalent to

$$C_{\text{basis}}(K) + 2rSK < 2dSK.$$

Subtracting $2rSK$ from both sides gives $C_{\text{basis}}(K) < 2S(d - r)K$, proving Eq. (9). When $C_{\text{basis}}(K) = C_{\text{basis}}$ is independent of K, $S > 0$ and $d - r > 0$, so division preserves the inequality and yields

$$K > \frac{C_{\text{basis}}}{2S(d - r)}.$$

Every algebraic step is reversible, proving both necessity and sufficiency. □

A.3 Proof for Proposition 4.3

Proof. At each SPSA update, the algorithm evaluates the scalar objective at the two parameter vectors $x_k + c_k\delta_k$ and $x_k - c_k\delta_k$. This count is two whether $x_k \in \mathbb{R}^d$ represents full coordinates or $x_k \in \mathbb{R}^r$ represents reduced coordinates. After S updates, either method has therefore used $2S$ scalar objective evaluations. Adding one final evaluation of the selected iterate gives $2S + 1$ evaluations per task and $(2S + 1)K$ evaluations across K tasks.

By definition, C_{basis} contains every additional forward circuit point used to construct a basis, evaluate a residual diagnostic, or refresh the basis. These points are not among the matched SPSA perturbation pairs or final objectives. Hence disjoint accounting gives

$$C_{\text{reduced}} = (2S + 1)K + C_{\text{basis}}, \qquad C_{\text{full}} = (2S + 1)K.$$

The assumed nonnegativity $C_{\text{basis}} \geq 0$ yields $C_{\text{reduced}} \geq C_{\text{full}}$. Subtracting the two expressions gives $C_{\text{reduced}} - C_{\text{full}} = C_{\text{basis}}$, so equality holds exactly when $C_{\text{basis}} = 0$. □

A.4 Proof for Theorem 5.1

Proof. Let $P = P^\top = P^2$ be any rank-r orthogonal projector. From $\nabla C_w = \mathbf{J}^\top w$,

$$\begin{aligned}
\mathbb{E}\left\|(\mathbf{I} - P)\mathbf{J}^\top w\right\|_2^2 &= \mathbb{E}\left[w^\top \mathbf{J}(\mathbf{I} - P)^\top (\mathbf{I} - P)\mathbf{J}^\top w\right] \\
&= \mathrm{tr}\big(\mathbf{J}(\mathbf{I} - P)\mathbf{J}^\top \mathbb{E}[ww^\top]\big) \\
&= \mathrm{tr}\big((\mathbf{I} - P)\mathbf{J}^\top LL^\top \mathbf{J}(\mathbf{I} - P)\big) \\
&= \mathrm{tr}\big((\mathbf{I} - P)AA^\top (\mathbf{I} - P)\big) \\
&= \|(\mathbf{I} - P)A\|_F^2 .
\end{aligned} \tag{15}$$

We used symmetry and idempotence in the second line and cyclic invariance of trace in the third.

Let $A = U\Sigma V^\top$ be a full singular-value decomposition with $U \in \mathbb{R}^{d\times d}$, and write the eigenvalues of $AA^\top$ as $\lambda_j = \sigma_j(A)^2$ in nonincreasing order. Since

$$\|(\mathbf{I} - P)A\|_F^2 = \operatorname{tr}(AA^\top) - \operatorname{tr}(PAA^\top),$$

minimizing the loss is equivalent to maximizing $\operatorname{tr}(PAA^\top)$ over rank-r projectors. Write $P = QQ^\top$ with $Q^\top Q = I_r$ and let $B = U^\top Q$. Then $B^\top B = I_r$ and

$$\operatorname{tr}(PAA^\top) = \operatorname{tr}\big(B^\top \operatorname{diag}(\lambda_1, \dots, \lambda_d)B\big) = \sum_{j=1}^{d} \lambda_j a_j,$$

where $a_j = \sum_{k=1}^{r} B_{jk}^2$, $0 \le a_j \le 1$, and $\sum_j a_j = r$. Because the λ_j are nonincreasing, a maximum of the linear functional is attained at $a_j = 1$ for $j \le r$ and $a_j = 0$ for $j > r$. This value is attained by $Q = U_r$, hence by $P_r = U_r U_r^\top$. The maximum retained energy is $\sum_{j\le r} \sigma_j^2$, so the minimum discarded energy is $\sum_{j>r} \sigma_j^2$. This proves the theorem, including the non-uniqueness when the boundary singular values are tied. □

B Algorithm

Algorithm 1: Matrix-free task-weighted QB

Given an anchor $\boldsymbol{\theta}$, training weights $\mathbf{W}$, tolerance ε, block size b, and maximum rank $r_{\max}$:

1. form the operator factor $L = K_{\mathrm{tr}}^{-1/2}\mathbf{W}^\top$ without forming $\widehat{\mathbf{M}}$;
2. expose $v \mapsto \mathbf{J}^\top L v$ and $u \mapsto L^\top \mathbf{J} u$;
3. draw a seeded Gaussian block Ω and compute $Y = A\Omega$;
4. orthogonalize Y twice against the recycled basis and the accumulated new basis;
5. append nondegenerate orthonormal columns and estimate the relative residual with independent probes;
6. stop at tolerance, rank exhaustion, or $r_{\max}$, recording the full status; otherwise draw a new block and repeat.

The two-pass orthogonalization controls numerical loss of orthogonality. The released status JSON records the rank and residual after every accepted block.

C Gradient-Free and Gradient-Based Methods

The legacy benchmark compares full ma-QAOA, symmetry tying, a dense fixed-rank oracle, two randomized tolerances, and an adjoint-free construction. Figure 7 separates forward observable calls, JVPs, VJPs, and dense Jacobian materializations. These are overlapping operator categories and must not be summed.

Gradient estimators on hardware can use parameter-shift identities [52, 62], while SPSA exchanges coordinate resolution for two noisy function values per step [56]. The released experiments do not equate these access models.

D Bias and Variance of Refresh Decisions

The diagnostic in Eq. (8) is a ratio of random quadratic forms. Even when its numerator and denominator are separately unbiased for scaled gradient energies under exact directional derivatives, the ratio is not unbiased. Finite differences add truncation error, and finite-shot objectives add measurement noise whose variance grows as the step shrinks.

A calibrated next-task refresh rule would require a labeled quantity such as the oracle benefit of rebuilding before that task. Split calibration could then be considered if calibration and future tasks were exchangeable [51]. Online drift would require a different guarantee [8].

E Repeated-Task Guidance Methods

E.1 No Refresh

One basis is built at the initial anchor and reused for all eight tasks. This minimizes construction but is exposed to anchor and task drift.

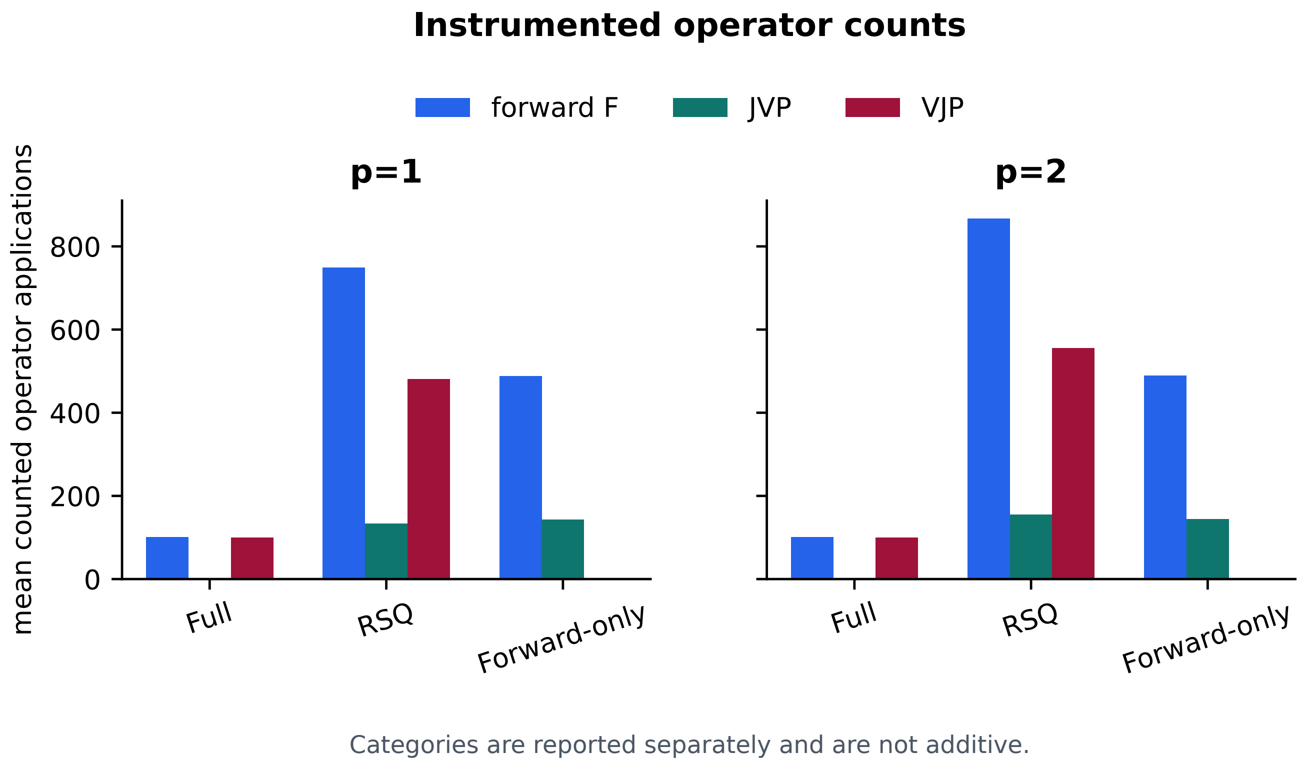

Figure 7: **Operator-access ledger.** Forward observable evaluations, finite-difference JVPs, simulator VJPs, and dense-Jacobian events are reported as distinct quantities.

E.2 Fixed Refresh

The basis is rebuilt after predetermined transitions. Timing is independent of observed residuals, which makes the policy a useful cost control.

E.3 Residual-Gated Refresh

Equation (8) is evaluated on eligible transitions. The tested threshold frequently triggers, so its cost approaches per-task refresh.

E.4 Per-Task Refresh

Every transition rebuilds from the current anchor with recycled initialization. This is the strongest local-adaptation control and the most construction-heavy.

E.5 Random Refresh

A seeded Bernoulli decision refreshes independently at each transition. It tests whether refresh count alone explains a gain.

E.6 Random Basis

A seeded isotropic rank-eight basis is fixed for the stream. It tests whether dimension reduction without learned sensitivity geometry is sufficient.

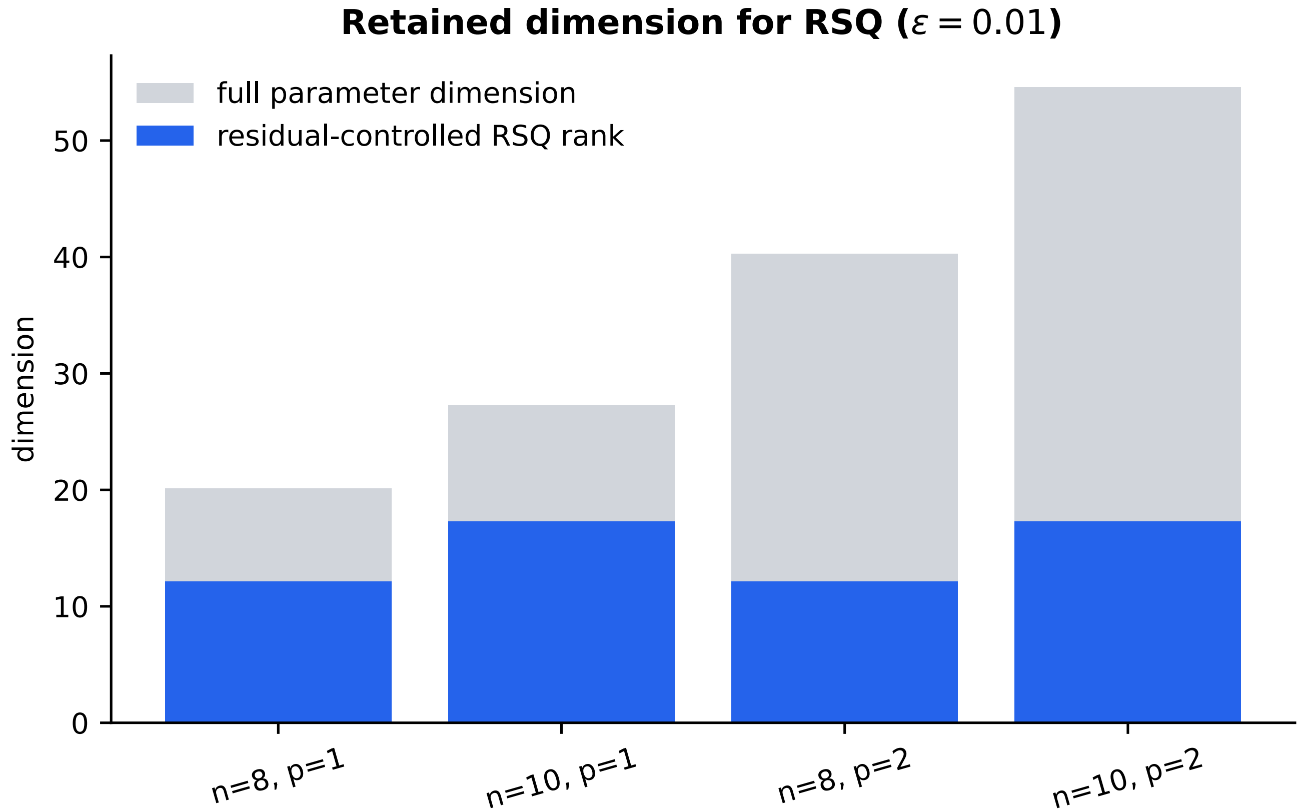

Figure 8: **Selected rank across the legacy grid.** Rank adapts to the local sensitivity spectrum but does not by itself predict end-to-end efficiency.

F Repeated-Task Benchmark Tasks

F.1 Regular Graphs

Random 3-regular graphs provide degree-homogeneous topologies at $n = 8$ and $n = 10$.

F.2 Erdős–Rényi Graphs

Edges are sampled independently with probability 1/2. Fixed graph seeds make the topology list reproducible.

F.3 Low-Rank Weight Drift

A positive base vector plus three orthonormal loadings defines the task family. An AR(1) latent state with persistence 0.85 and innovation scale 0.18 produces training and evaluation streams with independent realizations.

F.4 Independent Evaluation Stream

No evaluation weight enters $\widehat{\mathbf{M}}$. The shared base-and-loading law is the stated transfer assumption; performance outside that law is unmeasured.

F.5 Finite-Shot Objectives

Computational-basis samples estimate all edge observables jointly. The subspace path remains exact, so this task is a hybrid diagnostic rather than a device-native experiment.

Table 3: **Declared exact-development decision.** The residual-gated method passes the random-basis and forward-cost conditions but fails the one-percentage-point quality condition, so the conjunction fails.

Criterion	Required	Observed	Result
Quality versus full SPSA	≥ -0.010	-0.01567	**fail**
Quality versus random basis	> 0	+0.10048	pass
Forward-cost ratio versus full	≤ 1.25	1.21798	pass
Conjunctive development gate	all pass	observed	**fail**

Table 4: **Frozen repeated-task protocol.** Measurement repeats are nested before graph-level aggregation.

Track	Topologies	Depths	Tasks	Repeats	Evaluator
Exact development	8	2	8	1	exact statevector
Finite-shot pilot	8	1	8	3	256 shots

G Experimental Details for the Exact Benchmark

Full and reduced SPSA use $a_k = 0.12/(k+6)^{0.602}$, $c_k = 0.10/(k+1)^{0.101}$, and a maximum update norm of 0.40. The basis builder uses maximum rank eight, block size two, relative tolerance 0.10, six norm probes, and four residual probes. The refresh diagnostic uses one probe, step 0.01 in the exact track, step 0.10 in the shot track, and threshold 0.30.

H Experimental Details for the Finite-Shot Pilot

Each matched SPSA loop uses 81 optimization-objective points; residual-gated checks add separately counted objective points. Every such point uses 256 shots. The three measurement repeats share the graph, task, and optimization method but use independent measurement seeds. Exact post-evaluation of the selected candidate is used only for analysis.

I Physical Resource Accounting

Every row records task-local and cumulative counts for scalar objective points, vector-observable points, shots, JVPs, VJPs, builds, checks, refreshes, and their disjoint forward total. The ledger identities are validated before display generation.

The display manifest records and validates SHA-256 values for the legacy CSV, the exact and finite-shot task streams, and their three JSON summaries before any figure is regenerated. The audited paths are `experiments/results/maxcut_small.csv`, `experiments/results/amortized_development.csv`, and `experiments/results/amortized_shot_development.csv`, together with their corresponding JSON summaries.

Exact hash prefixes remain reproducible from `tables/table_reproduction.tex` and the display manifest. This integrity metadata is validated before rendering but is excluded from the five numbered experimental tables.

J Additional Analysis

The confirmatory design adds changepoints, independent task distributions, longer streams, larger graphs, readout perturbations, and device-native coordinate gradients. Classical-shadow ideas may reduce the cost of broad observable estimation in other operator families [30]; they are unnecessary for the commuting MaxCut measurements used here.

Table 5: **Finite-shot repeated-task audit.** "Shots" counts scalar objective shots; the VJP column records simulator-only construction.

Method	Ratio ± 2 s.e.	Δ full (points)	Forward	Shots	VJP	Refresh
Full SPSA	0.8387 ± 0.0376	0.00 (ref.)	648.0	165888	0.0	0.00
Residual gated	0.8353 ± 0.0392	-0.33 ± 1.67	771.7	173056	120.4	5.58
Per-task refresh	0.8448 ± 0.0345	$+0.61 \pm 1.57$	773.0	165888	150.9	7.00

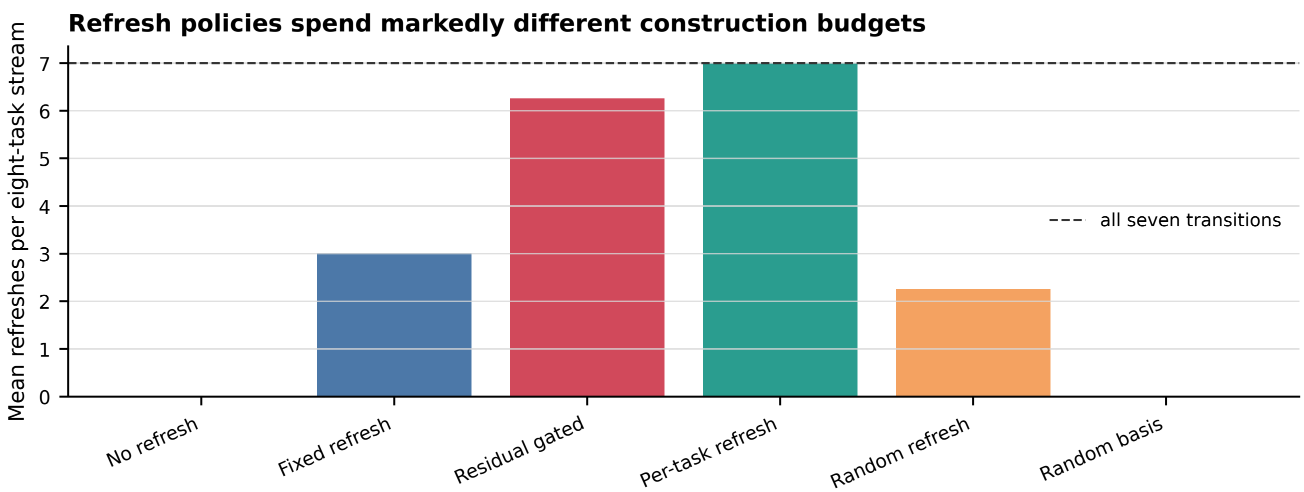

Figure 9: **Refresh burden.** The residual gate rebuilds on most of the seven transitions, whereas fixed and random controls spend smaller construction budgets.

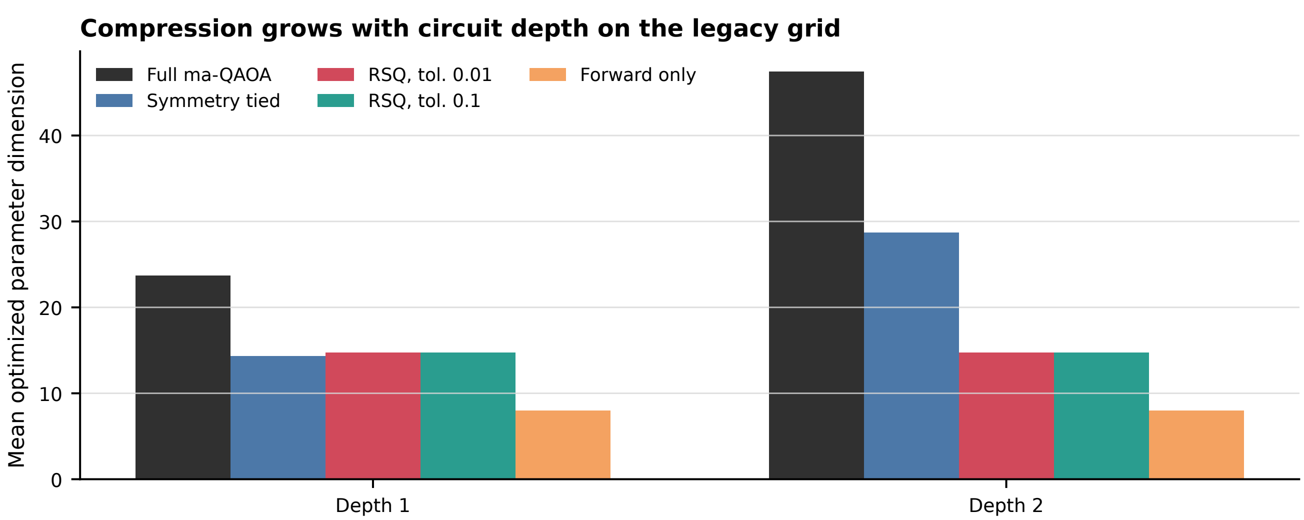

Figure 10: **Depth-dependent compression.** The number of retained directions grows more slowly than the ambient multi-angle dimension on this small legacy grid.

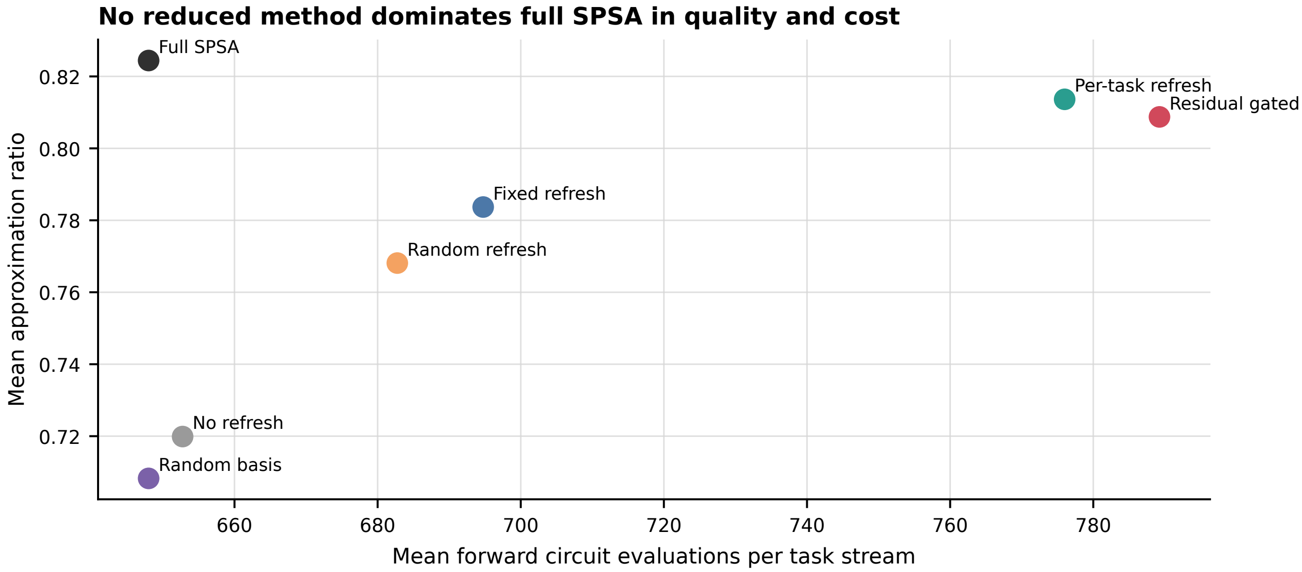

Figure 11: **Quality-cost frontier.** Mean forward cost includes scalar and vector-observable points. No reduced method simultaneously dominates full SPSA in the development audit.

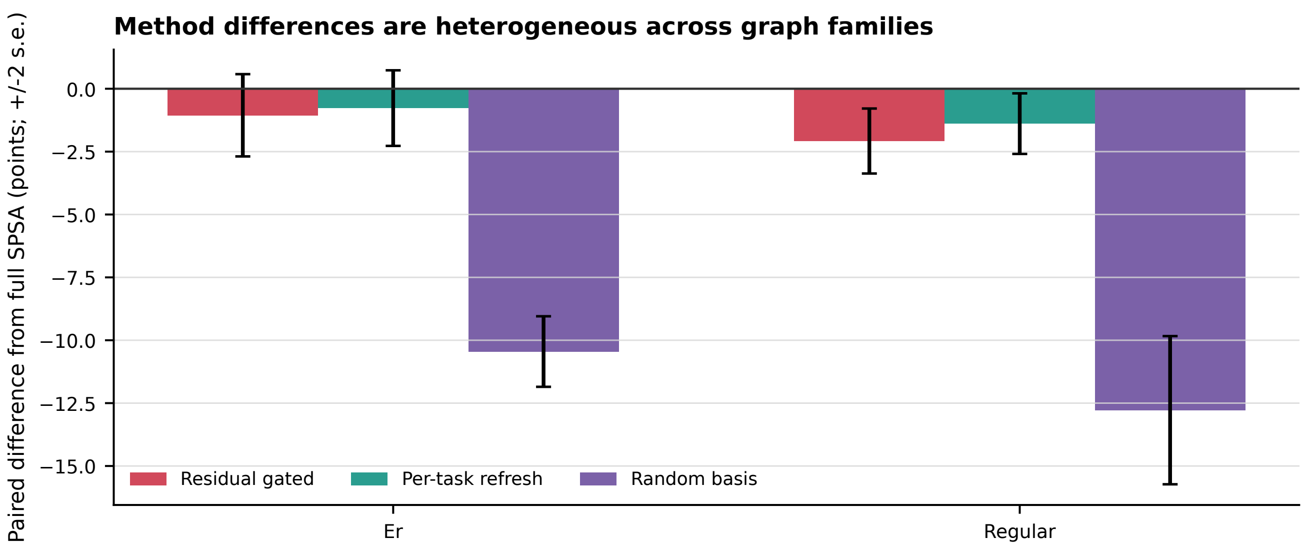

Figure 12: **Family heterogeneity.** Paired method differences vary across the configured graph generators; bars show descriptive ±2 standard errors.

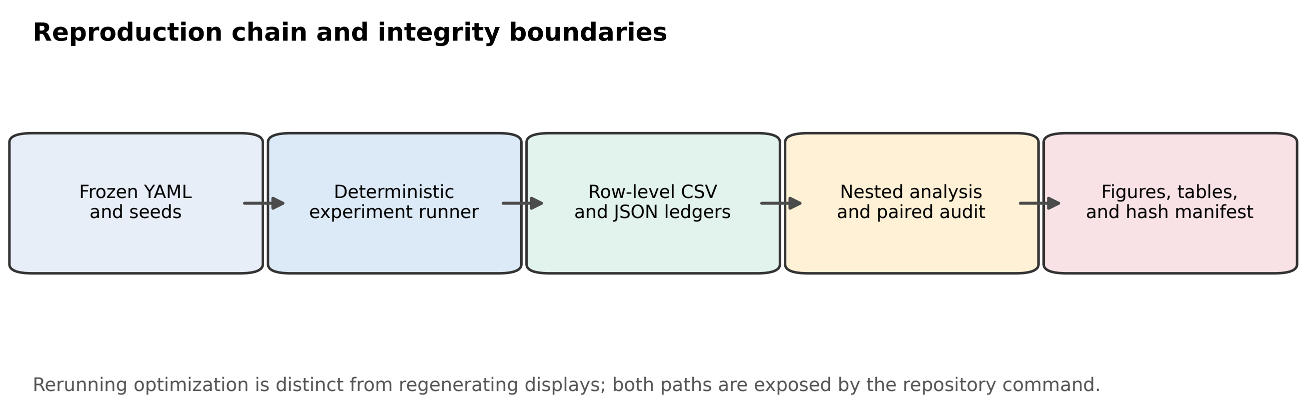

Figure 13: **Reproduction chain.** Frozen configurations and seeds produce row-level evidence; analysis creates displays and a hash manifest. Rerunning optimization and regenerating displays remain distinct operations.